

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



کابل پوهنتون

Microsoft Excel

استاد: پوهنیار عبدالله صافی

Introduction to MS Excel

- ❑ **Microsoft Excel** is a powerful **spreadsheet application** developed by **Microsoft**. It is used for organizing, analyzing, and storing data in tabular form.
 - Excel used to perform calculations, create charts, manage large datasets, and automate tasks using formulas and functions.
 - Excel is widely used in **business, education, finance, engineering, and research** because it helps users process data quickly and accurately.
- ❑ **Spreadsheet software** is a computer program used to organize, store, and calculate data in a table format made of **rows and columns**.

Continue....

- ❑ Excel arranges data in **rows** and **columns**, and each box is called a **cell**.
- ❑ Each cell can contain **numbers, text, formulas, or functions**.

Main uses of Excel

- ❑ **Data Organization:** Store and arrange data in rows and columns.
- ❑ **Calculations:** Perform automatic calculations using formulas and functions.
- ❑ **Data Analysis:** Analyze data using PivotTables, Sorting, and Filtering.
- ❑ **Data Visualization:** Create charts and graphs to represent data visually.
- ❑ **Reporting:** : Prepare professional reports
- ❑ **Dashboards:** A single page that displays important information using **charts**, tables, and key indicators
- ❑ **Statistical & Scientific Calculations:** Perform mathematical and statistical analysis.

Interface of MS Excel

- The **Excel Interface** refers to the environment you see when you open Excel. It contains several parts:
- **Title Bar:** Displays the name of the file and the application.
- **Ribbon:** A collection of tools and commands grouped in tabs (Home, Insert, Page Layout, etc.).
- **Quick Access Toolbar:** A small toolbar above the ribbon that provides shortcuts for common actions (Save, Undo, Redo).
- **Worksheet Area:** The main working space, divided into **rows** (horizontal) and **columns** (vertical).

Continue...

- **Formula Bar:** Displays the contents of the selected cell, and allows users to enter or edit data and formulas.
- **Status Bar:** Located at the bottom, showing information like page number, zoom, and calculation status.
- **Sheet Tabs:** Found at the bottom; allow users to switch between different worksheets in the workbook.

Ribbons in Excel

- The **Ribbon** is a toolbar at the top of the Excel window that organizes commands into tabs for easy access. Each tab contains **groups** of related commands.
- **Home Tab:** Basic formatting tools like font style, font size, text alignment, cut, copy, paste, and number formatting.
- **Insert Tab:** Tools for inserting charts, images, tables, pivot tables, and illustrations.
- **Page Layout Tab:** Options for adjusting margins, orientation, themes, and page setup for printing.

Ribbons in Excel

- **Formulas Tab:** Provides access to built-in formulas and functions (mathematical, statistical, logical, financial, etc.).
- **Data Tab:** Tools for sorting, filtering, importing data, and performing advanced data analysis.
- **Review Tab:** Proofreading tools like spell check, comments, and protection features.
- **View Tab:** Allows users to change workbook views, zoom settings, and show/hide gridlines or headings.

Workbooks in Excel

❑ A **Workbook** is the main Excel file that contains one or more worksheets.

✓ When you create a new Excel file, you are creating a workbook.

✓ A workbook can store multiple worksheets.

✓ Each workbook is saved with the extension **.xlsx**.

❑ Example:

A company may create one workbook for **Annual Sales Report**, and inside it there may be multiple worksheets for each month.

Worksheet

❑ **Worksheet:** A **Worksheet** is a single spreadsheet page inside a workbook.

- We can take 256 sheet in a single workbook.
- In a single worksheet 1048576 rows and 16384 columns.

❑ A worksheet contains:

- **Rows** (horizontal lines) قطارونه, ردیف‌ها
- **Columns** (vertical lines) ستون‌ها
- **Cells** (boxes where data is entered), cell The smallest unit of the worksheet, That box is called a Cell.

Cells, Rows, and Columns

❑ Cells:

- A cell is the intersection of a row and a column where data is entered.

❑ Rows:

- Rows run horizontally and are numbered (1, 2, 3, 4...).

❑ Columns:

- Columns run vertically and are labeled with letters (A, B, C, D...).
- Each cell has a unique address such as **A1**, **B5**, **C10**, etc.

Excel Working Environment

❑ The **Excel working environment** includes different tools and areas that help users create and manage spreadsheets.

❑ Important components include:

- Ribbon
- Formula Bar
- Name Box
- Worksheet area
- Sheet Tabs
- Status Bar

Formula Bar

❑ The **Formula Bar** is located above the worksheet.

It is used to:

- Enter formulas
- Edit data in a cell
- View the contents of the selected cell

For example, users can type formulas like:

=SUM(A1:A10)

This formula adds the values from cell A1 to A10.

Name Box

- The **Name Box** is located to the left of the Formula Bar.
- It shows the **address of the currently selected cell**.
- For example:
- If you click on cell **B3**, the Name Box will display **B3**.
- The Name Box can also be used to:
- Quickly navigate to a specific cell
- Assign names to ranges of cells

Key points about Workbooks:

- By default, a new workbook contains at least one worksheet.
- Users can add, delete, rename, or move worksheets within the workbook.
- All data, formulas, and charts are stored inside the workbook.
- A workbook can be saved in format **.xlsx**.

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



کابل پوهنتون

Microsoft Excel

استاد: پوهنیار عبدالله صافی

What is a Range in Excel

A **Range** in Excel refers to a group of selected cells.

Examples:

- A1:A5
- B2:D6
- C3:F3

Ranges are used for:

- Calculations, Formatting, Data analysis, Creating charts
- Selecting ranges allows users to apply actions to multiple cells at once.

Selecting a Single Cell

How to Select a Single Cell

- Steps:
- Move the mouse pointer to the desired cell.
- Click once on the cell.
- Example:
Clicking on **Cell A1** selects that cell.

Uses:

- Entering data
- Editing cell content
- Applying formatting

Selecting Multiple Cells (Range)

Selecting a Range of Cells

- Steps:
- Click on the first cell.
- Hold the **left mouse button**.
- Drag the cursor across the desired cells.
- Example:
Dragging from **A1 to A10** selects the range **A1:A10**.
- This method is called **Click and Drag Selection**.

Selecting Range Using Keyboard

Selecting Range with Keyboard

- Steps:
- Select the starting cell.
- Hold the **Shift key**.

Use **Arrow Keys** to expand the selection.

- Example:
- Select A1
- Hold **Shift + Right Arrow**

Advantages:

- Faster
- More precise

What is a Worksheet

A **Worksheet** is a page in an Excel workbook.

Each worksheet contains:

- Rows
- Columns
- Cells

Excel files can contain **multiple worksheets**.

Example:

- Sheet1
- Sheet2
- Sheet3

Moving Between Worksheets

Switching Between Sheets

- Steps:
- Click the sheet tab at the bottom.
- Select the sheet you want.
- Example:
Click **Sheet2** to open it.
- This allows users to organize data into different sheets.

Keyboard Shortcut for Sheet Navigation

Using Keyboard to Move Between Sheets

- Shortcuts:
- Move to next sheet
Ctrl + Page Down
- Move to previous sheet
Ctrl + Page Up

Benefits:

- Faster navigation
- Useful for large workbooks

Microsoft Account in Excel

Microsoft Account

- A Microsoft account allows users to:
- Save files to **OneDrive**
- Sync settings across devices
- Access Office online services
- Share files with others

Examples of Microsoft services:

- OneDrive
- Outlook
- Microsoft 365

Managing Microsoft Account

How to Manage Your Account

- Steps:
- Click **File**
- Select **Account**
- View account information

From here you can:

- Sign in
- Sign out
- Change user information
- Manage connected services

Quick Access Toolbar

What is Quick Access Toolbar (QAT)

- The **Quick Access Toolbar** is a customizable toolbar located at the top of Excel.

It provides quick access to commonly used commands such as:

- Save
- Undo
- Redo
- Print

Customizing Quick Access Toolbar

How to Customize QAT

- Steps:
- Click the **dropdown arrow** on the Quick Access Toolbar
- Select **More Commands**
- Add or remove commands

Examples of commands you can add:

- New
- Open
- Print Preview
- Sort
- Format Painter

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



کابل پوهنتون

Microsoft Excel

استاد: پوهنیار عبدالله صافی

Inserting Text

How to insert:

1. Click on the cell
2. Type your text
3. Press Enter to move down, Tab to move right

Examples:

- Enter Student Name in A1
- Enter City in B1

Inserting Numbers

How to insert: Click the cell → type the number → Enter

Examples:

- Enter 1200 as Sales amount
- Enter 3.5 for GPA

•Formatting:

- Home → Number group → Choose: Currency, Percentage, Decimal places

Inserting Dates

How to insert:

- 1.Type manually: 15-Mar-2026
 - 2.Use **Ctrl + ;** to insert the current date
 - 3.Excel formulas: =TODAY() → current date, =NOW() → current date and time
- Formatting:** Home → Number group → Short Date / Long Date
 - Example:** Enter project deadlines in a column

Editing Cells

How to edit:

- Double-click cell → type new content
- Or select cell → type → replaces content
- **Undo/Redo:** Ctrl + Z / Ctrl + Y
- **Find & Replace:** Ctrl + F / Ctrl + H for large spreadsheets
- **Example:** Correct a wrong product name or sales figure

Deleting Cells

- **Methods:**
 - Backspace → delete cell content
 - Right-click → Delete → Shift cells up/left
 - Select row/column → Right-click → Delete

Copy, Cut, and Paste Overview

- **Copy:** Duplicate content without removing original
- **Cut:** Move content to a new location
- **Paste:** Place copied or cut content in destination cells
- **Example:** Move sales figures from one sheet to another

Copy in Excel

- **How to copy:**
 - Select cell(s)
 - Ctrl + C or right-click → Copy
 - Select destination → Ctrl + V or right-click → Paste
- **Example:** Copy a column of student names to a new sheet
- **Tip:** Copy formulas with or without formatting using **Paste Special**

Cut in Excel

- **How to cut:**
 - Select cell(s)
 - Ctrl + X or right-click → Cut
 - Select destination → Paste
- **Use Case:** Move a row of data to another location
- **Tip:** Useful for reorganizing tables without retyping

Paste in Excel

- **Paste Options:**
 - Ctrl + V: Standard paste
 - **Paste Values:** Only numbers/text, not formulas
 - **Paste Formatting:** Apply formatting without changing data
 - **Paste Formulas:** Copy formulas only
- **Example:** Copy formulas from January to February sheet without overwriting formatting

Creating New Excel Files

- Steps:** File → New → Blank Workbook or Templates

- Tips:**

- Name files with meaningful titles
- Save immediately in desired location

- Example:** Create Budget_March2026.xlsx workbook

Opening Excel Files

- **Methods:**
 - File → Open → Browse
 - Double-click file in File Explorer
 - Recent Files → quickly reopen recent work
- **Example:** Open last month's sales report

Saving Excel Files

- **Steps:** Ctrl + S or File → Save
- **Options:**
 - Save locally or on **OneDrive**
 - Save as different formats: .xlsx, .xls, .csv, .pdf
- **Tip:** Use meaningful names and dates for versions

Moving and Copying Files

- Methods:**

- Drag & drop files in File Explorer
- Cut & Paste to move files to a folder
- Copy & Paste to duplicate files

- Example:** Move ProjectPlan.xlsx to project folder

Using OneDrive with Excel

- **Definition:** Cloud storage by Microsoft
- **Benefits:**
 - Access files anywhere
 - Automatic backup
 - Share with colleagues for collaboration
- **How to save:** File → Save As → OneDrive
- **Example:** Team working on shared Excel sales report

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



کابل پوهنتون

Microsoft Excel

استاد: پوهنیار عبدالله صافی

Introduction to File Saving

Saving a file means **storing your work permanently on the computer or cloud.**

Why saving files is important:

- Prevents **data loss**
- Allows **future editing**
- Helps organize documents
- Enables **sharing and collaboration**

Excel files normally use the extension:

.xlsx

Example:

Student_Grades.xlsx

Save Command

- The **Save command** stores the current workbook with the **same name and location**.
- **Shortcut Key:**
Ctrl + S
- How to Save:
- Click **File**
- Click **Save**
- Choose location (first time only)
- Click **Save**
- After the first save, Excel will **automatically update the same file**.

When to Use Save

You should use **Save** when:

- Updating an existing file
- Editing a previously saved document
- Adding new data to the same workbook
- Preventing work loss

Save As Command

The **Save As command** allows you to save a file:

- With a **different name**
- In a **different location**
- In a **different format**

Steps:

- Click **File**
- Click **Save As**
- Choose location
- Enter new file name
- Click **Save**

When to Use Save As

Use **Save As** when:

- Creating a **copy of a file**
- Changing **file format**
- Saving in a **new folder**
- Creating **different versions**

Example:

Original file:

Budget2025.xlsx

New file:

Budget2026.xlsx

File Formats in Save As

Format	Description
.xlsx	Standard Excel workbook
.xls	Older Excel format
.csv	Data format for databases
.pdf	Read-only document
.xlsm	Macro-enabled workbook

Saving Files to OneDrive

- Excel allows saving files to **OneDrive (Cloud Storage)**.
- Advantages:
 - Access files **anywhere**
 - Automatic **backup**
 - Easy **file sharing**
 - **Collaboration** with others
- Steps:
 - File → Save As → OneDrive

Introduction to Page Layout

The **Page Layout tab** controls how the worksheet will **appear when printed**.

- It allows you to adjust:
- Margins
- Orientation
- Page size
- Background
- Print area
- Page Layout helps make documents **professional and readable**.

Page Layout Tab Tools

Important tools in the Page Layout tab:

- Themes
- Margins
- Orientation
- Size
- Print Area
- Breaks
- Background

These tools help customize worksheet appearance.

Page Orientation

- Orientation determines the **direction of the page**.
- Two types:
- **Portrait**
- Vertical layout
- Best for documents and lists
- **Landscape**
- Horizontal layout
- Best for wide tables and reports
- Steps:
- Page Layout → Orientation

Page Margins

Margins are the **blank spaces around the edges of the page.**

Types:

- Top Margin
- Bottom Margin
- Left Margin
- Right Margin

Purpose:

- Improve readability
- Allow space for printing
- Make documents look professional

Introduction to Print Area

- A **Print Area** specifies which part of the worksheet will be printed.
- Without Print Area:
- Excel prints the **entire worksheet**.
- With Print Area:
- Excel prints **only selected cells**.

Setting Print Area

Steps to define Print Area:

- Select the cells to print
- Go to **Page Layout**
- Click **Print Area**
- Select **Set Print Area**
- Now only the selected section will print.

Clearing Print Area

- If you want to remove the print restriction:
- Steps:
- Page Layout → Print Area → **Clear Print Area**
- Excel will again print the **entire worksheet**.

Introduction to Protect Sheet

- Excel allows protecting worksheets to **prevent unauthorized changes**.
- Protection helps:
 - Prevent editing
 - Protect formulas
 - Secure important data
 - Control user access
- Common in **financial reports and student records**.

How to Protect a Sheet

Steps to protect a worksheet:

- Go to **Review Tab**
- Click **Protect Sheet**
- Enter a **password**
- Choose allowed actions
- Click **OK**
- Now users cannot modify locked cells.